

Lesson 6 — Build the Machine & Start the Motor

Matches LEGO Teacher Portal lesson: "Trunk Travels".

Objective (I can...): I can build the hardware machine and use code to start and stop one motor. **CS concept:** How hardware works — code controls a single motor (start/stop) · **Builds on:** Lesson 5 (we run ordered steps; now those steps drive a real part).

AI Connection (2 min, read aloud): Every smart robot has real parts — motors to move, sensors to notice. AI is the "thinking," but the motor is the "doing." *Ask the group:* "What jobs could a spinning motor do for a little robot?"

Materials: LEGO CS & AI kit + device per pair. **This is the unit's build lesson** — build the Hardware machine (hub + one motor) now and keep it assembled through Lesson 10.

60-minute flow - Warm-up (10): "Motors around us" — kids name things with motors (fan, toy car, blender) and spin their arms like a motor; on "stop!" they freeze. Start and stop, start and stop. -

Concept + demo (8): Build is coming — first show the Start motor and Stop motor picture blocks. Run one so the motor spins, press Stop so it halts. Point: the code is bossing a real part. - **Build/Code (25):** Pairs **build the Hardware machine** (the unit's one build — hub plus one motor), then code it to start the motor, wait, and stop. Try it a few times. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make the motor start, stop, then start again — a start-stop-start pattern. Guess before running. - **Share + tidy-up (10):** Pairs show starting and stopping and point to the motor. **Leave the machine built.**

Pair roles: Driver builds/codes, Navigator points to the motor and says "start... stop." Swap at the wiggle break. (*Trio rotates build/code/test.*)

Differentiation: *Support* — a palette card showing the start and stop pictures. *Extension / fast-finisher* — add a wait so the motor runs for a set time, then stops. *Reteach* — press Start and Stop by hand, naming each, before putting the blocks in.

Exit check: Pair's code reliably starts and stops the motor on purpose. **At-home (optional):** Find 2 things at home with a motor; tell a grown-up how you'd turn each on and off. ~5 min.
